# ANGULAR ASSIGNMENT : 03

## Module 3: Angular Components – Multiple Assignment Questions

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**Section 1: Component Basics**

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**1. Create your first component manually**

Create a component without using CLI and display simple text.
Hint: Create .ts, .html, .css files and register in module.

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**2. Create component using CLI**

Generate a component using Angular CLI.
Hint: Use ng generate component test.

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**3. Display data from component class**

Define variables in .ts file and display in HTML.
Hint: Use interpolation {{ }}.

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**4. Understand component structure**

Explore generated files and explain their purpose.
Hint: .ts, .html, .css, .spec.ts.

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**5. Use component inside another component**

Display child component inside parent component.
Hint: Use selector tag.

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**Section 2: @Component Decorator**

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**6. Explore @Component properties**

Identify and explain:

- selector
- templateUrl
- styleUrls

Hint: Check component .ts file.

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**7. Modify selector and observe changes**

Change selector name and update usage in HTML.
Hint: Selector acts like custom HTML tag.

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**8. Use template property instead of templateUrl**

Write inline HTML inside component.
Hint: Use template property.

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**9. Use styles property instead of styleUrls**

Write inline CSS inside component.
Hint: Use styles: [].

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**10. Compare inline vs external templates**

Create two components:

- One inline
- One external

Hint: Compare readability and maintainability.

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**Section 3: Component Interaction & Reusability**

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**11. Reuse component multiple times**

Use same component 3 times with different data.
Hint: Static data for now.

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**12. Create UI component (Card)**

Create reusable card component with:

- Title
- Description
- Button

Hint: Use multiple instances.

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**13. Nest components (Parent-Child)**

Create parent and child components and render child inside parent.
Hint: Use selector.

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**14. Organize components properly**

Create folder structure for components.
Hint: Group related components.

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**15. Create multiple components using CLI**

Generate:

- header
- footer
- sidebar

Hint: Use CLI multiple times.

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**Section 4: Component Lifecycle Overview**

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**16. Use ngOnInit lifecycle hook**

Log message when component loads.
Hint: Implement OnInit.

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**17. Observe component creation flow**

Add console logs in constructor and ngOnInit.
Hint: Check execution order.

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**18. Use ngOnDestroy hook**

Log message when component is destroyed.
Hint: Navigate between components.

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**19. Compare constructor vs ngOnInit**

Write example and observe behavior.
Hint: Initialization difference.

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**20. Track lifecycle during navigation**

Use routing and track lifecycle hooks on page change.
Hint: Console logs.

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**Section 5: Real World Practical Tasks**

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**21. Build simple dashboard layout**

Create components:

- Header
- Sidebar
- Content

Hint: Combine all in app component.

**22. Create profile component**

Display user profile details using component.
Hint: Name, email, city.

**23. Convert static HTML into Angular components**

Take static page and break into components.
Hint: Header, body, footer.

**24. Create reusable UI section**

Design section and reuse it multiple times.
Hint: Cards or list.

**25. Component-based UI design**

Design page using only components.
Hint: Think modular.

**Section 6: Interview Thinking Questions**

**26. What is a component in Angular?**

Explain with real example.

**27. What is the role of @Component decorator?**

Explain its importance.

**28. Difference between inline and external template**

Explain with use case.

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**29. Why components are reusable?**

Explain benefits.

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**30. What is lifecycle hook in Angular?**

Explain with example.

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